

SIDDARTH DAYASAGAR

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EDUCATION

Northeastern University, Boston, MA

Expected Dec 2026

Master's in Robotics (Electrical and Computer Engineering)

CGPA 3.63

Relevant Coursework:

Mobile Robotics, Robotics Mechanics and Control, Control Systems Engineering, Formal methods for dynamic systems, Legged Robotics, Pattern Recognition and Computer Vision

Jain University, Bangalore, India

Jun 2023

Bachelor of Technology in Robotics and Automation

CGPA 8.66

SKILLS

- **Programming:** Python, C++, MATLAB
- **Libraries:** ROS (noetic), OpenCV, PCL, ROS2 (humble, Jazzy), CasADi, Pinocchio
- **Design and Development Tools:** Fusion 360, Ultimaker CURA, Arduino UNO R3 and Raspberry Pi 4B+
- **Simulation:** Gazebo Classic, Gazebo harmonic, Mujoco, IsaacSim, IsaacLab
- **Certifications:** Udacity Robotics Nanodegree, Rigbetel Labs ROS Mentorship Program

WORK EXPERIENCE

Robotics Research Intern

Aug 2025 – Dec 2025

SpaceData Inc., Tokyo, Japan

- **Engineered** a PPO-based quadruped locomotion policy in Isaac Sim and **elevated** performance by approximately **70 %** over NVIDIA's pretrained baseline by **designing** improved rewards, **optimising** hyperparameters, and **implementing** domain randomisation.
- **Integrated** ROS 2 Nav2 with an MPPI controller to **establish** a robust indoor autonomy pipeline, **enabling** global planning, obstacle-aware local control, and multi-sensor navigation for disaster-response scenarios.
- **Pioneered** a next-generation outdoor navigation system by **building** a grid map-driven autonomy pipeline and **designing** an **MPC controller** that extends quadruped navigation from controlled indoor labs to large-scale, unstructured outdoor environments, unlocking multi-terrain deployment for disaster-response missions.
- **Architected Space Station OS**, a modular ROS 2 autonomy framework that **combined subsystems** into a unified control architecture, using behaviour trees, system diagnostics, and multi-node coordination to create a **scalable, fault-tolerant robotics simulation platform**.
- **Developed** physics-accurate simulations in Gazebo Harmonic and **integrated** them with a real-time PyQt operations dashboard, **synchronising** telemetry, controls, and diagnostics across distributed ROS 2 nodes, significantly **reducing integration time** and enabling rapid testing of autonomy algorithms.

PROJECTS

Proximal Policy Optimisation of Pick and Place using Panda Arm

Nov 2025 - Dec 2025

- **Trained** a PPO-based reinforcement learning policy to perform pick-and-place manipulation with a **7-DoF Franka Panda** arm in **MuJoCo**, using Cartesian end-effector actions for goal-directed control.
- **Implemented** training pipeline in **PyTorch** with Gaussian policies, GAE, clipped objectives, and entropy regularisation for stable learning in continuous action spaces.
- **Designed** task-specific reward shaping and safety penalties (reach-to-object, joint limits, smooth motion, collision avoidance) and validated pick-and-place performance using the MuJoCo viewer.

Autonomous Racing F1/10

May 2025- Sept 2025

- **Architected** a full-stack trajectory tracking and optimisation framework on the F1TENTH platform using **both Model Predictive Control (MPC) and Model Predictive Contouring Control (MPCC)** in ROS2 and CasADi
- Formulated a nonlinear kinematic bicycle model and embedded real-time constraints including velocity, steering angle, and curvature bounds; achieved stable control up to **4.5 m/s** on hairpin curves with less than **5% cross-track error**
- Integrated Signal Temporal Logic (STL) into the cost function to enforce formal safety rules (e.g., avoid boundary zones, reach goals), increasing safety robustness by **~28%** on average during lap simulations.

MPC-Based Missile GNC

Mar 2025- Apr 2025

- **Engineered** a high-fidelity missile guidance, navigation, and control (GNC) system using **Model Predictive Control (MPC) and Linear Quadratic Regulation (LQR)** in MATLAB with CasADi
- **Benchmarked** MPC's performance against LQR in constrained environments, showcasing **MPC's superior ability to handle state/input limits**, mitigate disturbances, and optimise manoeuvrability
- **Implemented** real-time constraint handling and predictive horizon tuning, **improving pitch stability by 45%** and **reducing control effort by 38%** compared to the baseline